

Paulo Eduardo dos Santos

SENIOR DATA ENGINEER | CLOUD & BIG DATA | DATA WAREHOUSE, DBT & BLOCKCHAIN DATA

ATS Match: 100.0% — Senior Data Engineer @ Consensys

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PROFESSIONAL SUMMARY

Senior Data Engineer with 9 years of experience designing, building, and maintaining robust **ETL/ELT data pipelines** in **Python**, **SQL**, and **Spark** for banks, payments, telecom, healthcare, and e-commerce organizations that process millions of daily transactions. Strong **SQL** skills paired with hands-on delivery on cloud **data warehouses** — **Snowflake**, **BigQuery**, and **Redshift** — and on **AWS** big-data services including **EMR / EMR Serverless**, **S3**, Glue, and Athena.

Hands-on across the modern data stack — **dbt** and **Airflow** for transformation and orchestration, **PySpark/Spark** for distributed processing, **Terraform** for infrastructure as code, and **CI/CD** automation to reduce manual work. Specialized in **Data Warehouse** and Data Lake / Lakehouse architecture (**dimensional modeling**, Star/Snowflake schemas, Medallion Bronze/Silver/Gold), with **data quality**, **data governance**, and security embedded into every pipeline. Proven results: 99.9% uptime SLA, 60% storage reduction, 50% faster processing, and 70% less manual operation. Builder of public open-source **blockchain data** projects (Ethereum, Solana, DeFi) that bridge strong enterprise data engineering with Web3/FinTech, and a collaborative partner to analysts and non-technical stakeholders.

TECHNICAL SKILLS

Core Languages: Python (advanced), SQL (advanced), PySpark, Shell Script, Scala

Distributed Processing & Big Data: Apache Spark, PySpark, SparkSQL, AWS EMR / EMR Serverless, Hadoop, Hive, Databricks, Trino

Transformation & Orchestration: dbt, Apache Airflow, Apache NiFi, Kafka, Talend, DataStage, SSIS, Informatica PowerCenter/BDM, Sqoop, Flume

Cloud Data Warehouses: Snowflake, BigQuery (GCP), Redshift (AWS), Synapse (Azure), Teradata

AWS: S3, EMR / EMR Serverless, Glue, Athena, Redshift, Lambda, Kinesis, Lake Formation, DMS, Step Functions, KMS, CloudWatch

Azure & GCP: Azure (Data Lake, Databricks, Data Factory, Synapse, Unity Catalog, Fabric) · GCP (BigQuery, Dataproc)

Architecture & Modeling: Data Warehouse (Star/Snowflake schema), Dimensional Modeling, Data Lake & Lakehouse (Medallion Bronze/Silver/Gold), CDC, Event-Driven, Cloud Migration (on-prem → cloud)

Lakehouse & Formats: Delta Lake, Apache Iceberg, Apache Hudi, Parquet

Data Quality & Governance: Data Quality frameworks, Data Governance, Data Catalog (AWS Glue, Unity Catalog), PII/LGPD compliance, HashiCorp Vault, AWS KMS, Fernet encryption

DevOps & IaC: Terraform, CI/CD, Docker, Kubernetes, Ansible, Git

Databases: SQL Server, Oracle, PostgreSQL, MySQL, MongoDB, Cassandra, HBase, DynamoDB

Visualization & Reporting: Power BI, QlikSense, Tableau, Spotfire, MicroStrategy, OBIEE, Grafana

Blockchain / Web3 Data: Dune Analytics, DeFi/NFT metrics, Ethereum & Solana on-chain data, cross-chain analytics

Methodologies: Agile for Analytics (Scrum, SAFe), requirements gathering for analytics projects

KEY ACHIEVEMENTS

- Sustained a **99.9% uptime SLA** for Banco do Brasil's Big Data environment processing millions of daily transactions
 - Cut maintenance time by **40%** via a reusable Python automation framework
 - Achieved **60% storage reduction** migrating legacy SQL tables to Parquet/Delta Lake
 - Reduced processing time by **50%** through Big Data cluster optimization at Claro Brasil
 - Reduced manual intervention by **70%** through Azure Automation pipelines
 - Led **multi-cloud migration** (Oracle → AWS + Azure + GCP) of Cielo payment data
 - Delivered a healthcare Data Lake on Apache Iceberg with full **PII/LGPD governance** (Vault, KMS, Fernet)
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PROFESSIONAL EXPERIENCE

Data Engineer — BRQ

Jul 2025 – Present | Remote, Brazil

RD Saúde — Healthcare Data Lake (Customer 360)

- Developed **PySpark/SparkSQL** ETL pipelines for large-scale ingestion, transformation, and consolidation of clinical data, with support for Apache Iceberg and Apache Hudi.
- Built a healthcare Data Lake with Medallion architecture (Bronze/Silver/Gold) on Apache Iceberg over **S3**, supporting electronic health records, scheduling, vaccines, medication, prescriptions, and tele-consultation use cases (Amplimed, HAOC, Vitat).
- Orchestrated **Airflow** DAGs integrated with **Spark** on Kubernetes, with automated YAML manifest generation through Python builders.
- Modeled the Customer 360 analytical table integrating multiple sources (CRM, scheduling, events, interests) served via Trino, Athena, and **Redshift**.
- Implemented an in-house **Data Quality** framework (cross-origin validations, light checks, DLS validations) and a YAML-versioned data catalog for **data governance**.
- Secured sensitive health data (PII/LGPD) with HashiCorp Vault, AWS KMS, and Fernet encryption, including decrypt pipelines in landing.
- Integrated API ingestion (Amplimed, HAOC, Precifica) with async S3 clients and evolving schema handling (JSON/Parquet/Iceberg).
- Built end-to-end observability (custom log handlers, Spark metrics, exception tracking, Iceberg table comments) integrated with AWS SNS and Step Functions; maintained Docker containers (dev/qa/prod) for PySpark and Python K8s jobs.

Itaú — Modernization & Analytics

- Developed **ETL** pipelines using **EMR** Serverless and Glue Studio; optimized **SQL** queries for performance and implemented **data quality** processes.
 - Performed A/B test analysis and developed metric-improvement analyses and hypothesis validation.
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Senior Big Data Specialist / Data Engineer — Stefanini

Jun 2023 – Jun 2025 | São Paulo, Brazil

Banco do Brasil — Big Data Environment Sustainment

- Built **dbt** models in **Snowflake** for financial reporting with **data quality** tests and versioned transformations.
- Sustained and optimized a production Cloudera/Hortonworks cluster, ensuring **99.9% uptime SLA**.
- Developed **Python** and Shell automation for operational processes; built Jupyter notebooks for exploratory analysis and Ansible jobs for infrastructure management.
- Optimized Hive queries and maintained HDFS structures.

AmericaNet — APIs & Data Processing

- Built **dbt/Snowflake** ELT models with integrity validations and Unity Catalog **data governance**.
- Developed a RESTful API in **Python** Flask to expose processed data; implemented complex transformations with

SparkSQL in Databricks.

- Automated processes with Azure Automation and built data pipelines with error handling and logging.

Cielo — Dashboard Analytics

- Developed **PySpark** scripts for high-volume processing and Athena external tables over **S3** with partitioning.
- Migrated analytical data to **Snowflake** with **dbt dimensional modeling** supporting Power BI dashboards.
- Analyzed and documented legacy Oracle procedures.

Itaú — ETL Process Refactoring

- Refactored legacy **SQL** tables to **Parquet** and converted stored procedures to **PySpark** jobs, achieving **60% storage reduction**.
- Built a data catalog with AWS Glue, implemented **data quality** processes, and produced detailed technical documentation of all migrated processes.

DirectLog — Geolocation System

- Built a pipeline capturing latitude/longitude via the Google Maps API and **Airflow** DAGs for batch address processing.
- Implemented an authenticated SFTP web interface, EC2 hosting, and a caching layer to optimize API calls.

Brasildev — AWS Data Lake

- Built a layered Data Lake (Bronze/Silver/Gold) with a data catalog and governance; created a **Python** ingestion engine with Salesforce integration.
 - Configured security/access policies with Lake Formation and automated infrastructure with **Terraform**.
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Senior Data Engineer — Natura & Co

Feb 2022 – Jun 2023 | São Paulo, Brazil

E-commerce Pipeline

- Modeled **dimensional** data for e-commerce analytics; developed **ETL** pipelines with **EMR Serverless** and **Airflow** DAGs for orchestration.
- Built **dbt** models in **Snowflake** for customer-behavior analysis with **data quality** tests.
- Built Delta Lake structures for data versioning and incremental load processes; integrated with BI via **Redshift**.

Cielo — Cloud Migration & Modernization

- Analyzed Oracle data structures and mapped them to **BigQuery**; migrated legacy data to Azure Blob Storage and built pipelines in Azure Data Factory with Unity Catalog cataloging.
- Converted complex **SQL** logic to distributed processing; developed **PySpark** and **SparkSQL** on Databricks; monitored spark-submit jobs.

JBS — Data Pipeline Modernization

- Provided technical leadership for a team of engineers and developers.
 - Implemented a high-availability NiFi cluster integrated with Active Directory; configured an Application Load Balancer and built NiFi flows for SAP ingestion and S3↔**Redshift** movement.
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Data Engineer — Enel

May 2021 – Feb 2022 | Brazil

Data Warehouse Migration

- Built a migration pipeline with Talend Open Studio moving data from Impala to **Redshift**.
 - Created QlikSense dashboards connected to **Redshift**, documented existing **ETL** processes, optimized queries, and trained business teams on the new tooling.
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Big Data Solutions Architect — Banco do Nordeste

Jan 2021 – May 2021 | Brazil

Corporate Data Lake

- Designed a Data Lake architecture with raw, stage, and analytics zones; implemented an unmanaged Hadoop cluster and configured Dremio for the analytics layer.

- Planned a NoSQL structure with Cassandra for a credit system; integrated legacy systems via DataStage and produced architecture documentation.

Bradesco — Data Ingestion Framework

- Updated ingestion-framework scripts for Hive and HBase tables; developed shell automation and integrated with the IBM IWS scheduler.
 - Connected to Teradata for extraction and optimized load/transformation processes.
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Big Data Architect / Engineer — Hospital Sírio-Libanês

Mar 2019 – Dec 2020 | São Paulo, Brazil

Hospital Analytics Platform

- Analyzed and ingested hospital data from multiple sources; installed and configured **Airflow** for orchestration and developed **ETL** DAGs.
- Implemented Docker containers integrated with Airflow, configured Kubernetes for scalability, integrated AWS Cognito with Azure AD for SSO, and set up a CloudWatch→Elasticsearch pipeline via Lambda.

Raízen — Data Ingestion Framework

- Surveyed the data-pipeline architecture and analyzed Azure Data Lake, Databricks, and Power BI; documented SAP integration via SSIS Theobald.

Dell EMC (Client: Fleury) — Consulting & Training

- Analyzed data and sustained a Hortonworks environment; configured security with Kerberos/Ranger and AD integration.
 - Developed monitoring scripts for Oozie jobs, built Grafana monitoring dashboards, and delivered operational training to the client team.
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Big Data Technical Lead — Claro Brasil

Sep 2017 – Mar 2019 | São Paulo, Brazil

Cloudera Cluster Implementation

- Installed and configured 3 Cloudera clusters for distinct projects (Data Lake, M2M, BI); implemented Kerberos with AD as KDC and cross-realm trust between clusters in different domains.
- Integrated Hive/Impala with BI tools (SAS, QlikView, Alteryx), built a data ingestion framework and monitoring/sustainment scripts, and integrated Informatica BDM for **ETL** — cutting processing time by **50%**. Mentored and led 2 engineers.

Nextel — Sustainment & ETL

- Developed **ETL** ingestion processes, analyzed data with Beeline/Impala, extracted data via Sqoop from relational sources, gathered functional requirements, and sustained the production environment.
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Big Data Analyst — Vivo

Jun 2017 – Sep 2017 | Brazil

Hadoop Migration & Sustainment

- Built **ETL** ingestion processes, analyzed data with Hive and **Python**, migrated Hadoop 2.2 → 2.6, persisted data in HBase, and managed the cluster with Zookeeper.
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EDUCATION

- **Associate Degree in Big Data and Data Intelligence** — Anhembí Morumbi University | Expected 2026
 - **Artificial Intelligence** (in progress) — FMU
 - **Bachelor of Science in Information Systems** — UNIP | 2006
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CERTIFICATIONS

- Red Hat Certified System Administrator (RHCSA)
 - Big Data Engineering — Garre Training (160h)
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OPEN-SOURCE PROJECTS — BLOCKCHAIN DATA & MODERN DATA STACK ([GITHUB.COM/PAULOEDUSPBR](https://github.com/pauloeduspbr))

- **dune-analytics-queries** — 12 **SQL** queries on Dune analyzing **blockchain data**: DeFi protocol metrics, NFT markets, Ethereum and Solana activity, and cross-chain comparisons.
 - **crypto-data-pipeline** — production-style pipeline ingesting market data for 100 crypto assets (CoinGecko API) with **Airflow**, PostgreSQL, and Docker; 7/7 automated **data quality** checks passing, 1.74s execution.
 - **medallion-data-lake** — production-ready Bronze/Silver/Gold template with **Airflow**, **dbt**, and **Terraform** (**AWS** + GCP), SCD Type 2 dimensions and 36 automated tests demonstrating **CI/CD**-ready, governed pipelines.
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LANGUAGES

- **Portuguese**: Native
- **English**: Upper-Intermediate (B2) — fluent technical reading and writing